

Smart Healthcare AI for ICD-10 Code Prediction from Unstructured Clinical Texts: A Comparative Study of Four Machine Learning Models

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Abstract—Accurate prediction of ICD-10 codes from unstructured clinical notes is essential for improving hospital operations, billing accuracy, and disease tracking. However, the complexity of clinical language and constraints in healthcare infrastructure pose significant challenges. This study evaluates four classical machine learning models—Naïve Bayes, Random Forest, Support Vector Machine (SVM), and XGBoost—for ICD-10 classification using clinical text data collected from four hospitals in Indonesia. The dataset comprises 92,000 annotated records and underwent standardized preprocessing and TF-IDF-based feature extraction. Models were assessed using classification metrics (accuracy, precision, recall, F1-score) and system performance indicators (latency, throughput, memory usage). XGBoost outperformed the others across most metrics, while Naïve Bayes proved effective in high-throughput scenarios. The results highlight the potential of lightweight models in real-world clinical environments, where interpretability and resource efficiency are as important as predictive performance. This study contributes to the advancement of smart healthcare AI and health informatics, particularly in the context of ICT adoption for resource-constrained hospitals in developing countries.

Keywords—ICD-10 classification, clinical text mining, machine learning, XGBoost, TF-IDF component, smart healthcare AI, health informatics, ICT for resource-constrained hospitals

I. INTRODUCTION

The International Classification of Diseases, 10th Revision (ICD-10) is a standard that is used all around the world to classify medical conditions, treatments, and diagnoses. Accurate ICD-10 coding is important for more than just keeping track of patients in today's healthcare systems. It is also critical for billing, reporting to public health, and data-based analytics. But a lot of healthcare facilities, especially those with few resources, still do ICD-10 coding by hand. Competent coders have to read and sort through free-text diagnoses that doctors write by hand. This can lead to errors, slow down processing, and make medical information systems work harder [1].

Thanks to improvements in natural language processing (NLP) and machine learning (ML), computerized ICD-10 coding is now a possible and exciting option. Prior research has shown that standard machine learning methods, including SVM [2], Naïve Bayes [3], Random Forest [8], and XGBoost [4], may be utilized to classify clinical text. People like these algorithms better since they are faster, easier to understand, and don't need as much computer power as deep learning methods.

Conversely, the majority of preceding investigations have concentrated on English datasets and utilized structured data from Electronic Health Record (EHR) systems in high-income nations [5]. There is a significant deficiency of research evaluating the efficacy of ICD-10 categorization models on Indonesian clinical data, which is frequently compiled informally, has acronyms, and may comprise a blend of local languages and medical terminology. It's not viable to use existing models directly in the Indonesian healthcare system because the language and context are different.

Healthcare organizations also need organized ICD-10 data to help them decide how to keep an eye on diseases, figure out what to do when there is an outbreak, plan for buying new equipment, hire new staff, and develop new services. There is a tremendous need for computationally efficient models that may be used in places with lower resources, like Indonesia, where many health facilities still have insufficient IT infrastructure.

Improvements in large language models (LLMs) and deep learning architectures have helped ICD-10 prediction testing function better [6]. But it can't be used in real hospitals because it needs a lot of gear and has latency problems. This study assesses the effectiveness of four lightweight machine learning models—Naïve Bayes, Random Forest, SVM, and XGBoost—in the classification of ICD-10 utilizing unstructured diagnosis information in Bahasa Indonesia.

This initiative aims to solve the subsequent research inadequacies:

- **Linguistic Representation:** Most research don't look at the features of Indonesian-language clinical

writing, which is less structured and more varied than English electronic health records (EHRs).

- **Comparative Model Assessment:** A lot of the research just looks at one algorithm. There is a lack of empirical comparisons across several lightweight ML models utilizing Indonesian clinical diagnosis data.
- **Unstructured Input:** There aren't many studies that look at how classic models deal with clinical inputs that are brief, loud, and full of abbreviations, which are common in frontline healthcare activities.
- **Practical Deployment:** There hasn't been enough testing of inference speed and memory usage, which are critical things to think about when putting models into real-time healthcare systems with limited computing power.

This paper examines the inadequacies that facilitate the deployment of automated ICD-10 coding systems in a realistic and scalable manner, in accordance with the linguistic, infrastructural, and operational needs of healthcare professionals in Indonesia and comparable areas.

II. LITERATURE REVIEW

Predicting ICD-10 codes from clinical text rapidly evolved in medical NLP. Researchers have studied rule-based systems, models, and deep learning architectures such as BioBERT demonstrated promising results at the cost of high computational resources [6].

Despite advances in deep learning, classic models such as SVM and XGBoost remain relevant and suitable for use, especially when combined with the use of TF-IDF. In healthcare, the ability to understand the model is crucial, as clinicians must understand the reasoning behind the predictions. Therefore, conventional models are still widely and relevantly used, especially in developing countries.

A. Support Vector Machine

SVM is effective in high-dimensional spaces, making it suitable for the complex task of medical code prediction from clinical notes, which often contain numerous potential codes. The combination of TF-IDF and SVM has been shown to yield competitive results, with precision metrics indicating a strong performance in predicting relevant codes [2].

B. Naïve Bayes

In the classification prediction of ICD-10 codes, Naïve Bayes offered several advantages when compared to other machine learning methods. It is simple, efficient, and robust, making it a suitable tool to predict medical data classification. Computation of this model is efficient either for quick training or prediction, which is crucial when handling large datasets. [3].

C. Random Forest

Random Forest algorithms are outstanding at classifying data, which can help make ICD-10 coding in medical diagnosis much more accurate. This machine learning method can work with complicated datasets, which makes it a good fit for the complicated nature of medical coding. Random Forests employ decision tree ensemble techniques and bootstrap aggregation to avoid overfitting and make models

more flexible. This is critical for accurate ICD-10 coding. The algorithm works much better for improving diagnostic coding since it can deal with missing data and complicated data. This will help doctors make more accurate and dependable diagnoses [8].

D. XGBoost

XGBoost is a complex machine learning technique that has been employed in a number of healthcare solutions, like automating ICD-10 coding. This method makes it easier and more accurate to transfer unstructured data into structured ICD-10 codes. This is very useful and vital for public health analytics. The XGBoost model is a substantial improvement from older machine learning methods like decision trees, support vector machines, and logistic regression. [4].

III. MATERIAL & METHODS

The approach utilized in this study is organized into five principal phases: dataset, preprocessing, feature extraction, modeling, and evaluation, as outlined in Figure 1.

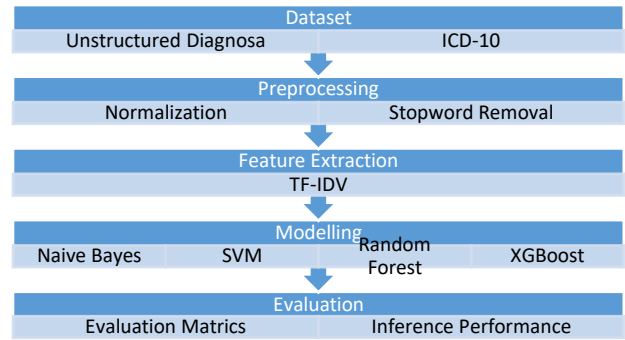


Fig. 1. Methodology of Study.

A. Data Set

The dataset was collected from the electronic medical record systems of four private hospitals in Indonesia, comprising a total of 92,000 records of unstructured clinical notes. The data was specifically sampled from the top 12 most frequent ICD-10 diagnosis codes in each hospital to ensure a representative distribution of common clinical conditions. The distribution of data is as follows:

- 1) Hospital A: 16,500 records.
- 2) Hospital B : 22,000 records.
- 3) Hospital C: 28,400 records.
- 4) Hospital D : 25,100 records.

Each clinical note was manually annotated with ICD-10 codes by certified medical coders. The text data underwent preprocessing steps including unnecessary word and stop word removal, and transformation using TF-IDF to convert the unstructured text into numerical feature representations suitable for machine learning models.

B. Preprocessing

A collection of hospital clinical reports, including patient medical records, will be assembled as part of the solution and methodically connected to the relevant ICD-10 diagnosis codes. The primary goal of the preparation process is to

normalize the text by lowering capitalization, deleting punctuation and numeric symbols, and getting rid of terms like "the" and "and." To further improve the dataset's quality, medical-oriented stopwords like "mg" and "patient" are expelled.

C. Feature Extraction

This study uses TF-IDF as the main way to extract features before training classification models because it strikes a good mix between efficacy and efficiency. TF-IDF, or Term Frequency-Inverse Document Frequency, is a means to give terms a number based on how important they are in a document relative to the full corpus.

$$\text{TF-IDF}(t, d) = \frac{\text{count}(t, d)}{\text{total words in } d} \times \log \frac{N}{\text{total docs contain } t}$$

Fig. 2. TF-IDF Formula.

D. Models Evaluated & Server Specification

This study employs a pragmatic and cost-effective methodology tailored for resource-limited hospital settings by evaluating four established models: Naïve Bayes, Random Forest, SVM, and XGBoost. Evaluation takes into account more than just how well the classification works. It also looks at technical performance metrics, like how quickly it can come to a conclusion and how well it consumes memory. These are particularly critical for making sure that clinical systems can talk to each other in real time.

1) *Naïve Bayes*: Used as a starting point, presuming that the features are independent.

2) *Support Vector Machine*: Used for multi-class classification with a linear kernel.

3) *Random Forest*: A group of decision trees that work together to lower model variance.

4) *XGBoost*: A gradient boosting model that is noted for being fast and effective.

Experiments were performed on PythonAnywhere (free tier), which offers limited resources (1 shared CPU core, 512 MB RAM), effectively simulating the constraints faced by numerous healthcare facilities.

E. Evaluation

Models were evaluated on a 20% test split using two categories of metrics:

a) *Classification Metrics*: Accuracy, Precision, Recall, and F1-Score.

b) *System Performance Metrics*: Average latency, peak memory usage, and throughput (samples per second).

IV. RESULT ANALYSIS

A. Analysis Based on Classification Metrics

Naïve Bayes delivered moderate performance, consistent with its simplicity. SVM achieved high precision but slightly lower recall than Random Forest. Random Forest showed consistently high classification performance. XGBoost achieved the best results across nearly all classification metrics, demonstrating its ability to handle complex feature interactions.

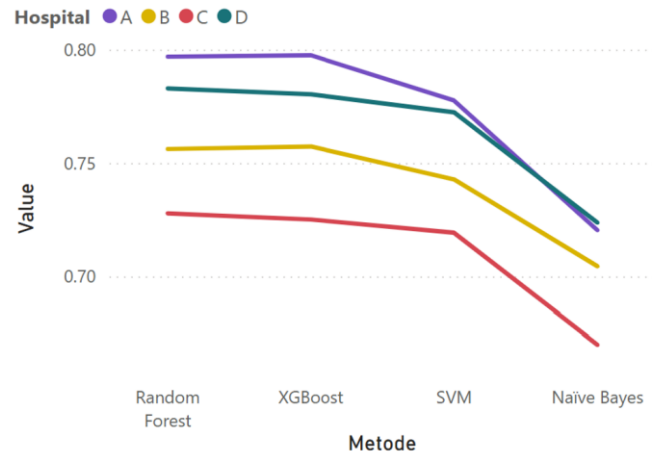


Fig. 3. Classification - Accuracy Comparison Between Models.

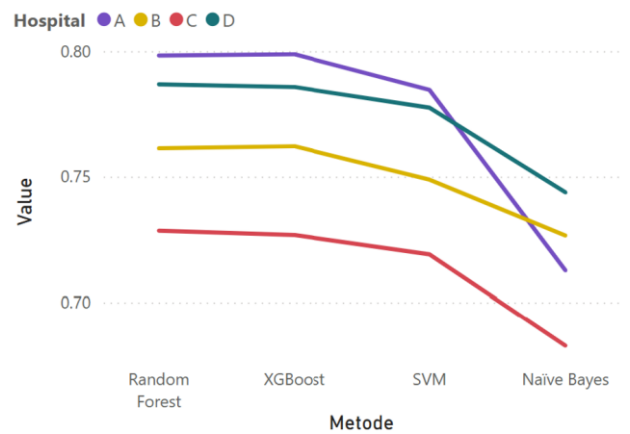


Fig. 4. Classification - Precision Comparison Between Models.

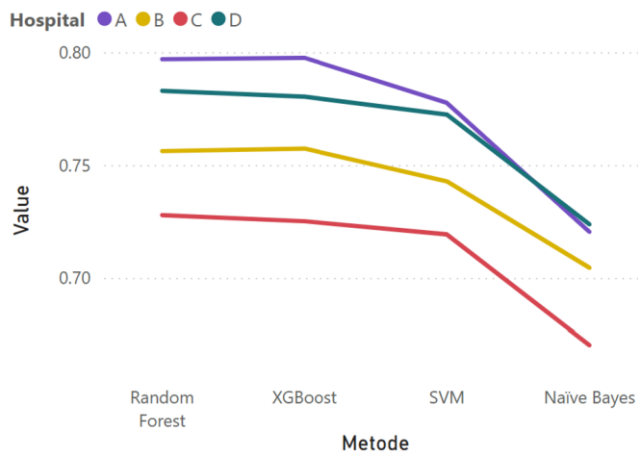


Fig. 5. Classification - Recall Comparison Between Models.

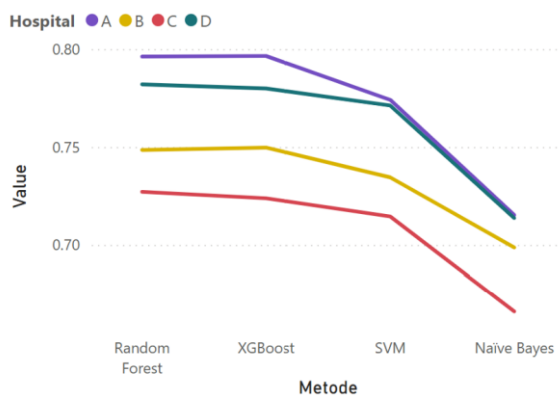


Fig. 6. Classification – F1 Comparison Between Models.

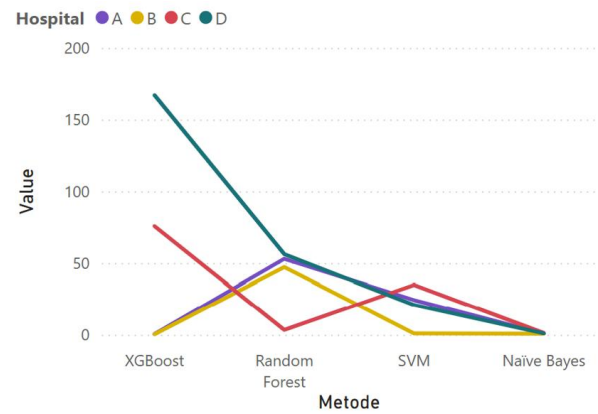


Fig. 7. Average Latency Per Sample (in ms) Comparison.

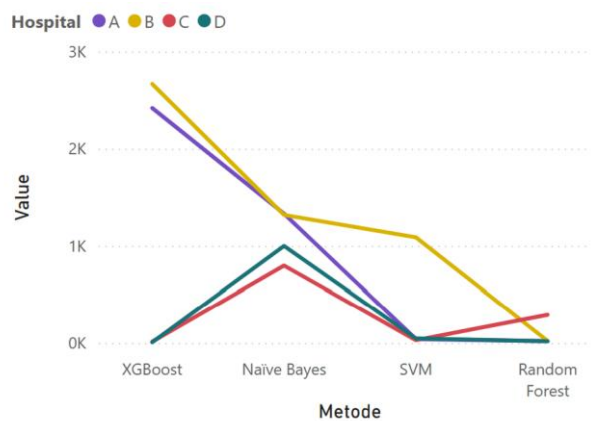


Fig. 8. Throughput (samples/sec) Comparison

B. Analysis Based on System Performance Metrics

Naïve Bayes excelled in throughput, making it suitable for real-time processing. Random Forest had high latency and low throughput, making it less ideal for online systems. SVM offered a balance between latency and accuracy. XGBoost outperformed all models in system performance: fastest, most memory-efficient, and highest throughput—ideal for deployment on edge devices or clinical cloud systems.

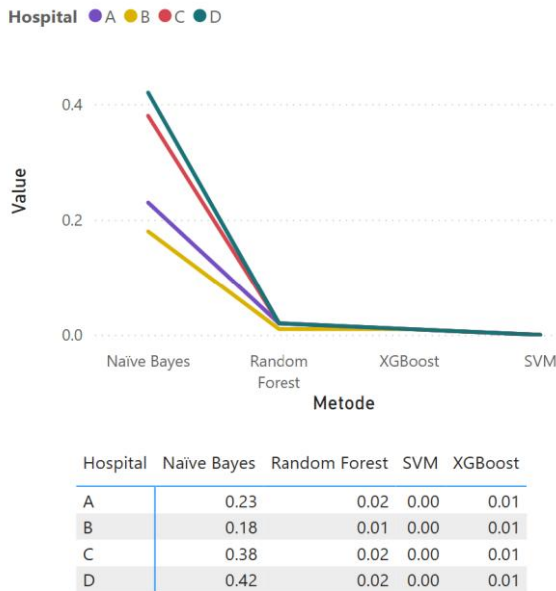


Fig. 9. Peak Memory Usage (MB) Comparison

V. CONCLUSION

This study conducted a comprehensive evaluation of four classical machine learning models—Naïve Bayes, Random Forest, SVM, and XGBoost—for predicting ICD-10 codes from unstructured clinical notes in Indonesian hospitals. By analyzing both classification metrics and system performance (latency, throughput, memory), the study addressed the practical challenges of deploying predictive models in resource-constrained healthcare environments.

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The dataset used in this study has been anonymized and is available for academic and non-commercial use at the following location: <https://github.com/antonprevianto/ICD-10>. The data includes free-text medical diagnoses written in Bahasa Indonesia along with manually labeled ICD-10 codes. Any identifiable patient information has been removed in accordance with ethical standards and applicable data protection regulations.

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